

Predicting Total Medicare Beneficiaries by State (2023)

By: Team G11 (Sidharth Dusanapudi, Ashish Sunkarapalli,
Jayden Sincere Brown, Krish Venkatram Mallina)

Introduction

Total medicare beneficiaries serves as a measure of a demographic's healthcare need, and government officials can use this information to apportion resources to hire a sufficient number of healthcare workers. We will build a model that predicts total medicare beneficiaries per US state based on multiple factors chosen through reputable primary and secondary sources. If successful, this model will be vital in accurately predicting the healthcare needs of states, allowing lawmakers to designate the correct resources towards the healthcare sector each year.

1. Do states with higher average age and a Democratic political affiliation correlate to having fewer total Medicare beneficiaries?
2. Do states with greater healthcare provider availability (physicians per 100,000) have higher total Medicare beneficiary counts?
3. Do states with higher median family incomes have lower total Medicare beneficiary counts?

Data Summary

Metadata Summary

The response variable and the state-level Medicare summaries were taken from the 2023 CMS Medicare Geographic Variation files, which provide the beneficiary count, average age, HCC score, and percent male for each state. Additional state characteristics were merged from recent public sources: median family income from Statista/Census reporting, educational attainment from the American Community Survey, physician availability from Becker's ASC Review, and political affiliation from recent electoral patterns. The final analytic data set contains one 2023 observation for each of the 50 states, so the unit of analysis is the state rather than the individual beneficiary.

Several variable manipulations were needed before modeling. Education, physician availability, and total state population were converted to quartiles so that the qualitative predictors had interpretable four-level groupings, matching the project requirements. The total-state-population entries for AK, AL, AR, and AZ were also realigned to their correct states before the population quartiles were created. The main reliability concern is that the file combines several sources that were collected with different procedures, and the quartile coding simplifies the original numeric measures. Even with that limitation, the merged data set is appropriate for a state-level exploratory regression analysis because all variables describe the same general 2023 setting.

EDA Summary

This report only uses one 2023 observation for each of the 50 states. Before calculating the population quartiles, the total-state-population values for AK, AL, AR, and AZ were adjusted to be consistent with the listed state abbreviations provided in the original data set. The response variable, total Medicare beneficiaries, is strongly right-skewed: counts range between 120,280 and 7,080,940, with mean of 1,360,708, a median of 994,359, and IQR of 1,132,502. Because the mean is well above the median and the strong influence of a small number of extremely large states dominating the upper tail, a log transformation of the response variable seems appropriate before fitting the final regression model.

Among the quantitative predictors, HCC score has the clearest positive association with beneficiary totals, with Pearson correlation 0.562 on the untransformed response variable and 0.702 when using $\log(Y)$. Percent male shows the strongest negative correlation of the quantitative predictors, with correlations -0.28 on the untransformed response variable and -0.467 when using $\log(Y)$. Average age appears weak on its own ($r = -0.038$ on the raw scale), which makes the political affiliation-age interaction worth investigating rather than relying on a simple main-effect interpretation. Median income is nearly unrelated to the raw response variable ($r = 0.051$) and only very weakly negatively correlated when using $\log(Y)$ ($r = -0.019$), so any income effect may depend on adjustment for other predictors.

In terms of qualitative predictors, population quartile exhibits the strongest visual separation, where the fourth-quartile average is at 2,360,061 whereas in the first quartile it is only 860,492. Political affiliation also shows some distinctions in means of the number of beneficiaries, which for Swing and Democratic states is higher compared to Republican states. On the other hand, the relationship between beneficiary totals and categorical predictors such as education and physician quartile is somewhat less consistent, as the average number of beneficiaries increases from Q1 (which is 819,413) to Q3 (2,083,766) before declining in Q4. When assessing potential interactions using $\log(Y)$, the relationship between average beneficiary age and log-transformed beneficiaries varies heavily depending on political affiliation. While Republican and swing states exhibit relatively flat trends, Democratic states display a steep negative association with $\log(Y)$, pointing to a strong interaction effect that war-

rants inclusion in the final model. Other investigated interactions, including HCC score by physician quartile and median income by population quartile plotted against $\log(Y)$, yielded nearly parallel trend lines, indicating no substantial interactive effects among those specific explanatory variables.

Analysis and Results

The response variable was modeled on the log scale rather than the raw count scale. This choice was supported both by the right-skewed histogram in Appendix C and by a Box-Cox profile on the raw response, which gave an estimated transformation parameter of 0 with a support interval of $[-0.2, 0.2]$. Because 0 falls in that interval, a log transformation is justified and also makes the residual diagnostics noticeably more stable. After transforming the response, the quantitative predictors used in interaction terms were centered so that the main effects were easier to interpret and the collinearity introduced by interaction terms was reduced.

Model building started with an additive model containing all eight explanatory variables, followed by formal tests of the three interaction ideas motivated by Appendix C. The age-by-political-affiliation interaction significantly improved the additive model ($F = 5.825$, $p = 0.007$), so the first research question is supported by the regression analysis. In contrast, HCC score by physician quartile ($F = 0.22$, $p = 0.882$) and median income by population quartile ($F = 1.056$, $p = 0.382$) did not improve model fit, so those interaction ideas were not kept. Appendix D therefore begins with the full interaction model that retained age by political affiliation and all remaining main effects.

From that full interaction model, terms were removed one at a time using nested F tests. Centered percent male was removed first ($p = 0.374$), followed by centered median income ($p = 0.558$), and then physician quartile ($p = 0.636$). None of these removals significantly worsened model fit, and the resulting model sequence in Appendix D shows steadily improving adjusted R^2 , AIC, and PRESS RMSE as the weaker terms were dropped. This stepwise elimination path leads directly to the final model with centered age, political affiliation, the age-by-political-affiliation interaction, centered HCC score, education quartile, and population quartile.

The final model explains a substantial amount of the state-to-state variation in Medicare beneficiaries, with adjusted $R^2 = 0.679$ and overall $p = <0.001$. HCC score is the strongest quantitative predictor in the final model ($p = <0.001$), and higher population quartiles continue to be associated with larger beneficiary totals, especially Q4 versus Q1 ($p = 0.005$). The age effect depends on political affiliation: for the Democratic reference group, higher average age is associated with a lower log beneficiary total, while the Republican interaction term significantly offsets that slope. This directly answers the first research question by

showing that age does not have one common effect across political groups. The second and third research questions receive weaker support. After adjustment for the rest of the model, physician availability is not a significant contributor, and median income does not explain additional variation beyond the retained predictors.

The multicollinearity and residual checks for the final model were acceptable. The largest VIF in Appendix D is 5.629, which indicates moderate but not severe collinearity after centering the interaction term. The residual standard error is 0.56 and the PRESS RMSE is 0.627, so predictive error stays in the same general range even under leave-one-out style validation. In the diagnostic plots, the residual-versus-fitted and scale-location panels do not show a strong funnel pattern, and the normal Q-Q plot is reasonably straight. The largest Cook’s distance is 0.114, so no single state appears to dominate the fit, although a few states do have relatively high leverage because the model includes several categorical terms and an interaction. Overall, the model is not perfect, but it is useful for describing broad state-level patterns in Medicare beneficiaries and for answering the project research questions.

Conclusions

The final regression shows that the most meaningful state-level signals are medical risk, population size, and the interaction between average age and political affiliation. Using the reduced model,

$$\begin{aligned}\widehat{\log(\text{Beneficiaries})} = & 13.158 - 0.261(\text{centered age}) + 9.197(\text{centered HCC}) \\ & + \text{education terms} + \text{population terms} \\ & + \text{political-affiliation adjustments} + \text{age-by-party adjustments}.\end{aligned}$$

For Democratic states, each one-year increase in average age above the sample mean is associated with about a 23% decrease in expected beneficiaries, holding the other retained predictors constant. For Republican states, the age slope changes by the interaction estimate, so the net age effect is much flatter and slightly positive. HCC score is also practically important: a 0.01-point increase in HCC is associated with an estimated 9.6% increase in expected beneficiaries, holding the remaining predictors fixed. In addition, states in the highest population quartile have about a 108.2% higher expected beneficiary total than comparable Q1 states. As one example of prediction, for Florida’s covariate values the model predicts about 4,856,324 Medicare beneficiaries, compared with the observed 5,250,841.

Taken back to the research questions, the analysis supports the idea that age matters differently across political-affiliation groups, but it does not support strong independent effects for physician availability or median income once other variables are controlled. The most important limitations are the small sample size of 50 states, the fact that all conclusions are

observational and cross-sectional, and the loss of detail created by converting some predictors to quartiles. Future work could improve the analysis by using multiple years of data, incorporating a direct measure of healthcare capacity beyond physician counts, and validating the model on a larger regional data set such as counties instead of states.

Appendix A: Data Dictionary

Data dictionary for the cleaned analysis dataset.

Variable	Abbreviation	Description
State	State	Two-letter state abbreviation.
Beneficiaries	Benef	Total Medicare beneficiaries in the state (measured in people).
Avg_Age	Age	Average beneficiary age (measured in years).
HCC_Score	HCC	Hierarchical condition category risk score (unitless index/multiplier).
Pct_Male	PctMale	Proportion of beneficiaries who are male (measured from 0 to 1).
Median_Income	Income	Median annual family income (measured in US Dollars).
Political_Affiliation	Party	State political affiliation. Levels: Democrat, Republican, Swing.
Education_Q	EduQ	Quartile of bachelor's degree percentage by state. Levels: Q1 Low, Q2, Q3, Q4 High.
Physician_Q	PhysQ	Quartile of physicians per 100,000 residents. Levels: Q1 Low, Q2, Q3, Q4 High.
Population_Q	PopQ	Quartile of corrected total state population. Levels: Q1 Low, Q2, Q3, Q4 High.

Appendix B: Data Rows

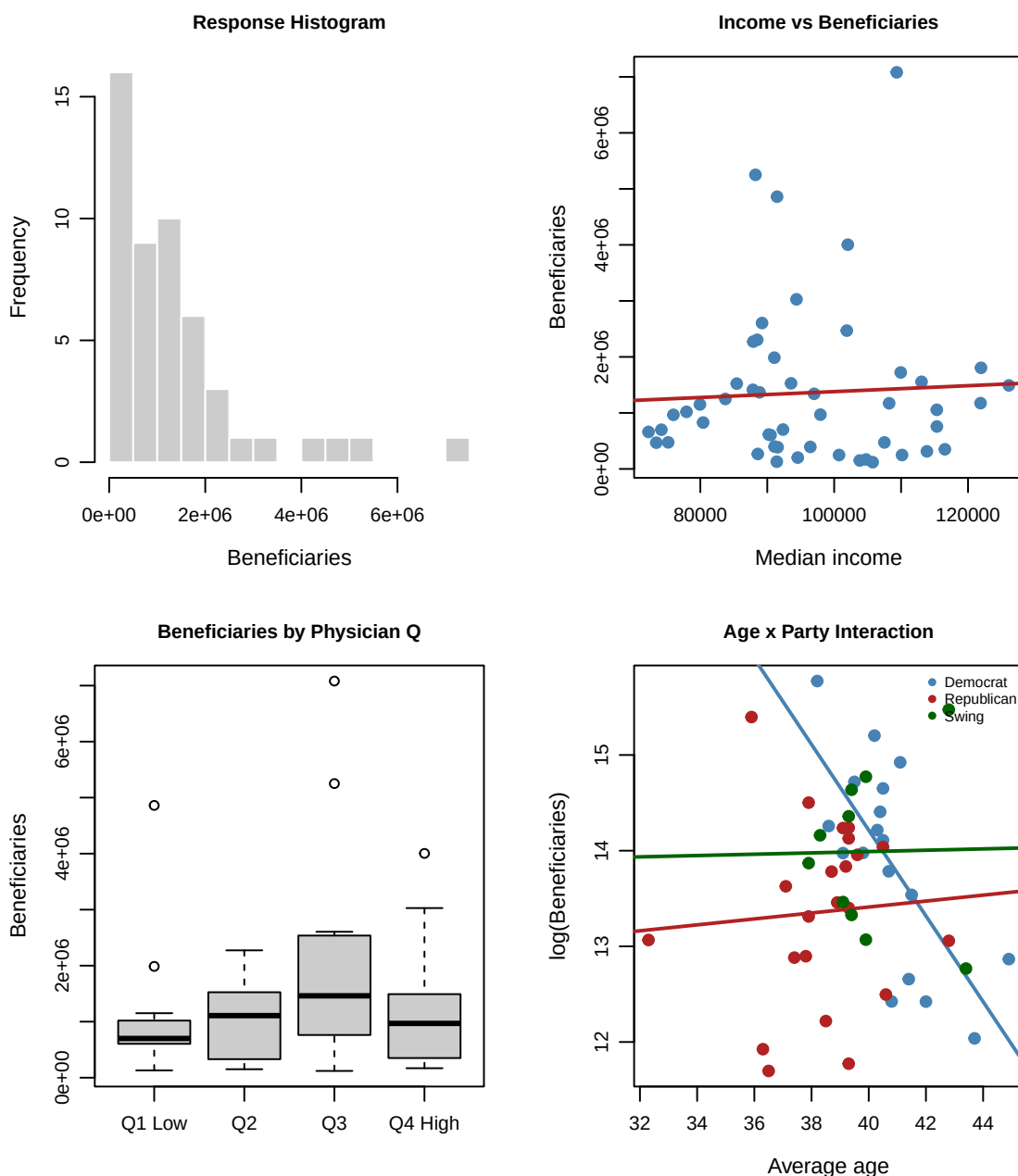
head() of the cleaned analysis data.

	State	Beneficiaries	Avg_Age	HCC_Score	Pct_Male	Median_Income	
1	AK	120280	36.5	0.82	0.4918	105752	
2	AL	1151244	39.6	1.02	0.4535	79940	
3	AR	699823	38.9	1.00	0.4531	74179	
4	AZ	1528056	39.3	0.92	0.4635	93553	
5	CA	7080940	38.2	1.04	0.4567	109349	
6	CO	1055744	37.9	0.89	0.4708	115335	
	Political_Affiliation	Education_Q	Physician_Q	Population_Q	log_Beneficiaries		
1	Republican	Q2	Q3	Q1 Low	11.69758		
2	Republican	Q1 Low	Q1 Low	Q3	13.95635		
3	Republican	Q1 Low	Q1 Low	Q2	13.45858		
4	Republican	Q2	Q2	Q3	14.23951		
5	Democrat	Q3	Q3	Q4 High	15.77292		
6	Swing	Q4 High	Q3	Q3	13.86976		

Note: output comes out in 2 rows (from 1-6) due to page width constraints. These may be considered as one continuous row.

Appendix C: Exploratory Data Analysis

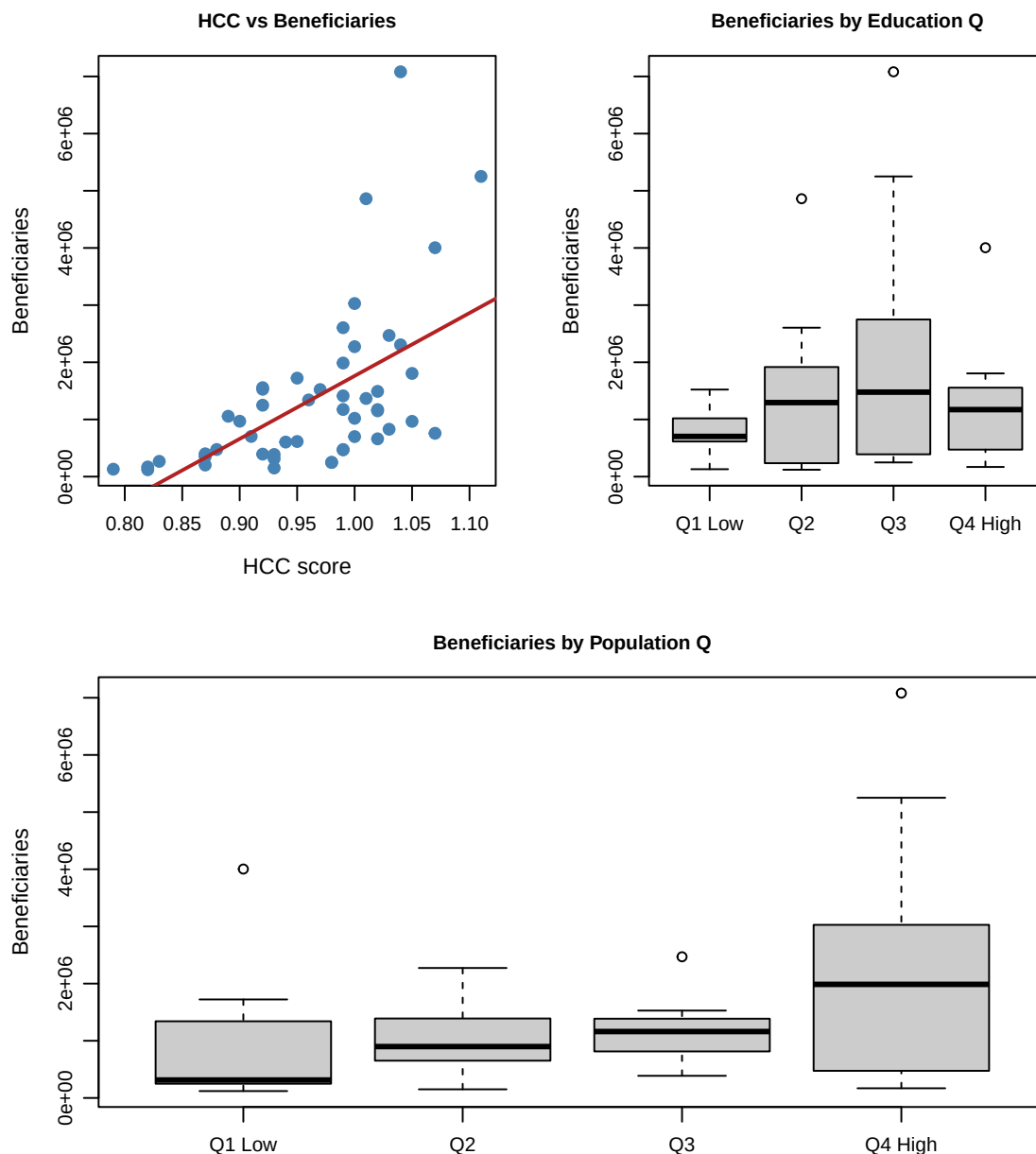
This appendix is limited to seven graphical summaries so that it stays within the two-page guideline while still covering the response, each research question, and the strongest supporting patterns.



On this page, the response histogram shows the strong right-skew that motivated modeling $\log(Y)$ rather than the raw count. The age-by-political-affiliation panel addresses the first research question and shows clearly nonparallel trends, especially between Democratic and Republican states. The physician-quartile boxplot and income scatter address the second

and third research questions. Both show weaker unadjusted patterns, which is why those effects needed formal regression testing rather than visual interpretation alone.

Taken together, these four visuals explain the overall modeling strategy. The response plot justifies transforming Y , the interaction screen motivates the only interaction retained in the final model, and the physician and income panels show why those two research questions needed to be answered with adjusted regression tests instead of raw visual differences alone.



The second page focuses on the strongest supporting predictors kept in the final analysis. HCC score has the clearest positive quantitative association with beneficiary totals, with raw-scale correlation 0.562, and population quartile shows the strongest qualitative separation, with the highest quartile averaging 2,360,061 beneficiaries compared with 860,492 in Q1. Education quartile shows some separation as well, though the pattern is less monotone than population, which matches the more modest education effects in the final regression.

These supporting plots also help explain why the final reduced model kept HCC score, education quartile, and population quartile. HCC score contributes a clear quantitative trend, population quartile captures a large structural difference in state size, and education quartile provides a smaller but still useful qualitative adjustment that improves fit after the

major predictors are accounted for.

Appendix D: Final Model Output and Plots

This appendix is organized as a backward-elimination sequence. It begins with the full interaction model, `log(Beneficiaries) ~ centered age * Political Affiliation + centered HCC score + centered percent male + centered median income + Education Quartile + Physician Quartile + Population Quartile`, and then removes one nonsignificant term at a time until reaching the final reduced model.

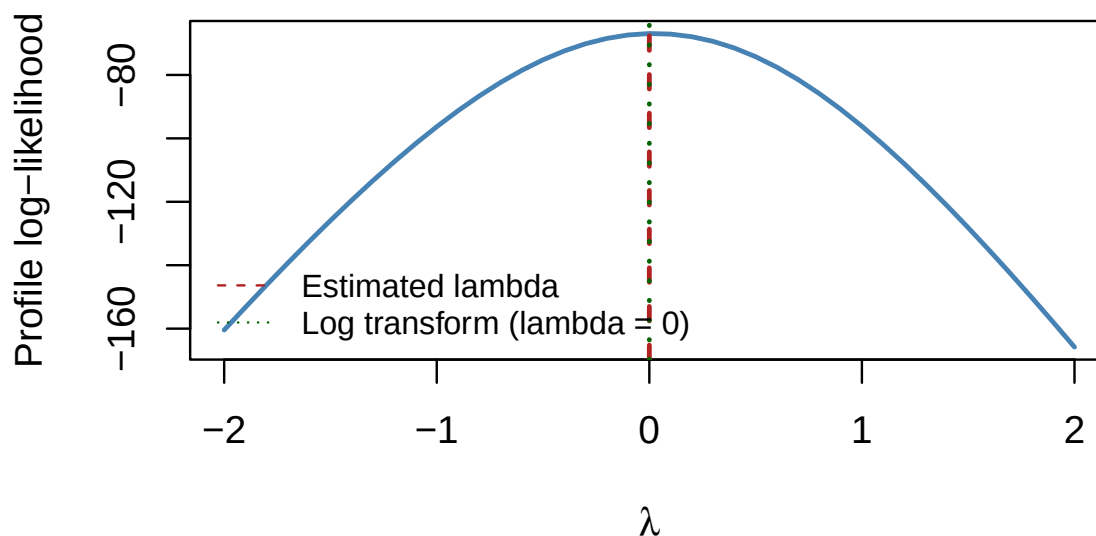
The order of output here mirrors that elimination process. Each step includes the most relevant visual or numeric evidence for that stage, so the graphs are embedded throughout the workflow rather than being clustered only at the end.

Step 1: Confirm the response transformation

Technique	Estimated_Lambda	Support_Interval	Interpretation
Box-Cox profile on the raw response	0	[-0.2, 0.2]	Lambda near 0 supports a log transformation.

The Box-Cox result is shown first because it justifies transforming the response before interpreting any coefficient or elimination table. Since the supported interval contains 0, the later model-reduction steps are carried out on a log-transformed response for principled rather than cosmetic reasons.

Box-Cox Profile for the Response

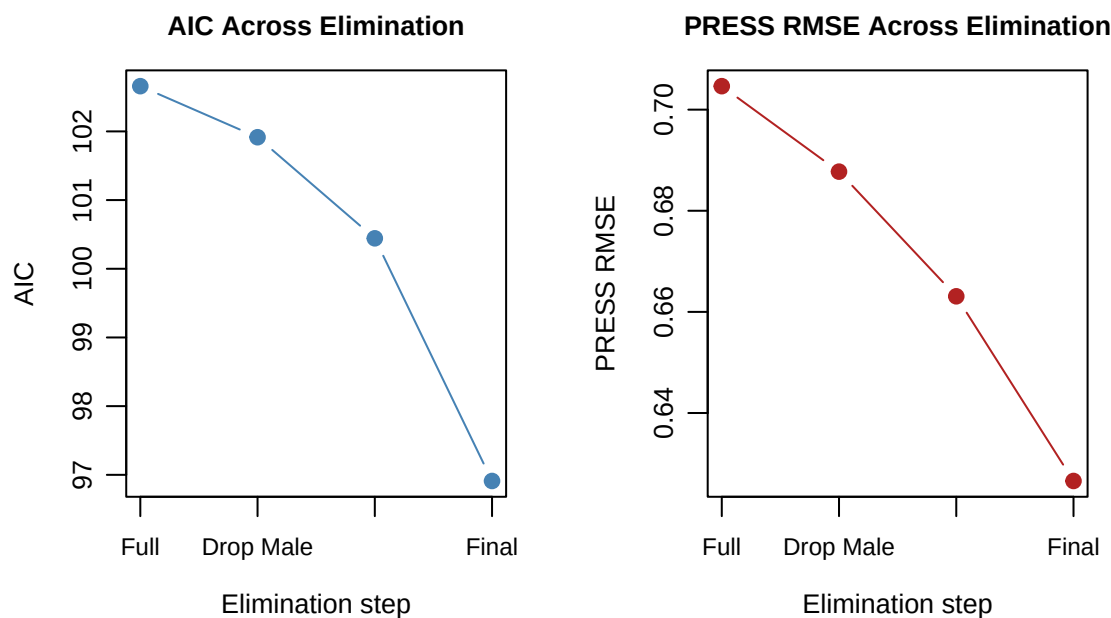


The profile plot makes the same transformation decision visually. The peak occurs near 0, and the vertical reference lines show that the chosen log transformation sits right where the profile is strongest.

Step 2: Start with the full model and remove weak terms

Stage	P_Value	Adj_R2	AIC	PRESS_RMSE
Full interaction model	–	0.659	102.7	0.705
Drop centered percent male	0.374	0.661	101.9	0.688
Drop centered median income	0.558	0.668	100.4	0.663
Drop physician quartile	0.636	0.679	96.9	0.627

This elimination table is the core of the Appendix D workflow. Starting from the full model, centered percent male was dropped first, then centered median income, and then physician quartile. Each removal has a large p-value, so none of these terms contributes enough adjusted explanatory value to offset the improvement in parsimony, and the fit criteria move in the right direction as the model becomes smaller.

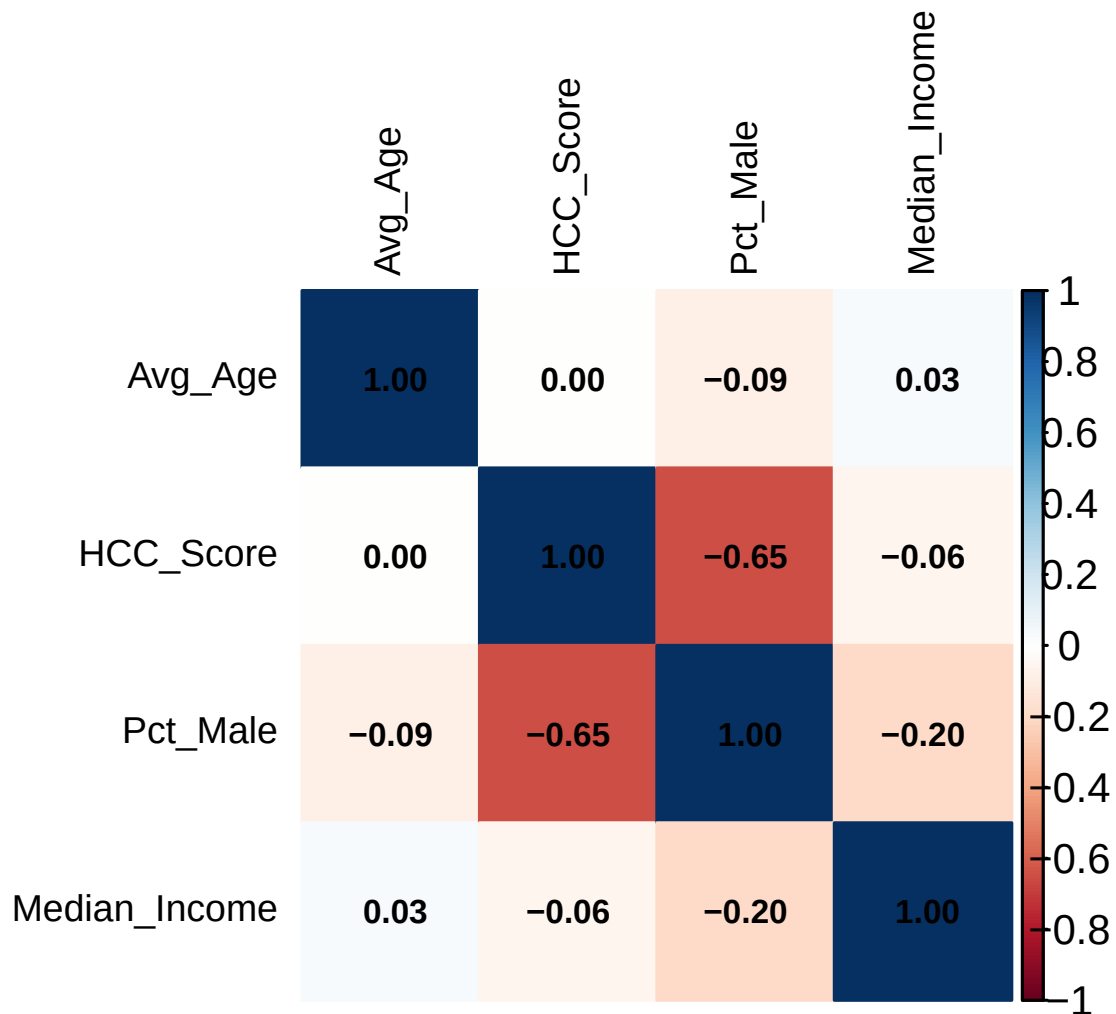


These two plots show the same elimination story graphically. Both AIC and PRESS RMSE improve as the weaker terms are removed, so the simpler final model is not only easier to interpret but also more stable by the criteria used in this report.

Step 3: Examine the full model before simplification

	Term	Estimate	Std_Error	t_value	P_Value
(Intercept)	Intercept (Dem, Edu Q1, Pop Q1)	13.280	0.475	27.972	<0.001
c_Avg_Age	Centered age	-0.327	0.115	-2.849	0.008
Political_AffiliationRepublican	Republican	-0.438	0.466	-0.941	0.354
Political_AffiliationSwing	Swing	0.197	0.383	0.514	0.611
c_HCC_Score	Centered HCC score	7.745	2.066	3.749	<0.001
c_Pct_Male	Centered percent male	-9.288	10.293	-0.902	0.374
c_Median_Income	Centered median income	0.000	0.000	-0.773	0.445
Education_QQ2	Education Q2	0.884	0.338	2.616	0.013
Education_QQ3	Education Q3	0.876	0.347	2.523	0.017
Education_QQ4_High	Education Q4 High	0.614	0.489	1.255	0.218
Physician_QQ2	Physician Q2	-0.163	0.290	-0.563	0.577
Physician_QQ3	Physician Q3	-0.287	0.405	-0.709	0.483
Physician_QQ4_High	Physician Q4 High	-0.044	0.573	-0.077	0.939
Population_QQ2	Population Q2	0.060	0.272	0.221	0.827
Population_QQ3	Population Q3	0.388	0.270	1.439	0.160
Population_QQ4_High	Population Q4 High	0.684	0.279	2.455	0.020
c_Avg_Age:Political_AffiliationRepublican	Age x Republican	0.401	0.123	3.261	0.003
c_Avg_Age:Political_AffiliationSwing	Age x Swing	0.079	0.171	0.461	0.648

The full-model coefficient table gives the starting point for backward elimination. In this fuller specification, several terms are already weak individually, especially centered percent male, centered median income, and the physician quartile indicators, which is consistent with the later decision to drop them from the model.



The quantitative-predictor correlation plot provides a first visual screen for possible redundancy among the original numeric predictors before the final model is interpreted. None of the pairwise relationships is large enough to suggest severe redundancy by itself, but it still makes sense to follow this with formal VIF checks later in the appendix.

Step 4: Summarize the final reduced model

	Term	Estimate	Std_Error	t_value	P_Value
(Intercept)	Intercept (Dem, Edu Q1, Pop Q1)	13.158	0.346	37.981	<0.001
c_Avg_Age	Centered age	-0.261	0.089	-2.926	0.006
Political_AffiliationRepublican	Republican	-0.168	0.335	-0.503	0.618
Political_AffiliationSwing	Swing	0.284	0.291	0.976	0.335
c_HCC_Score	Centered HCC score	9.197	1.378	6.674	<0.001
Education_QQ2	Education Q2	0.606	0.249	2.427	0.020
Education_QQ3	Education Q3	0.695	0.281	2.476	0.018
Education_QQ4 High	Education Q4 High	0.397	0.307	1.293	0.204
Population_QQ2	Population Q2	0.082	0.255	0.322	0.749
Population_QQ3	Population Q3	0.360	0.254	1.420	0.164
Population_QQ4 High	Population Q4 High	0.733	0.243	3.015	0.005
c_Avg_Age:Political_AffiliationRepublican	Age x Republican	0.348	0.111	3.149	0.003
c_Avg_Age:Political_AffiliationSwing	Age x Swing	0.010	0.156	0.063	0.950

The final-model coefficient table gives the estimates after the elimination process is complete. Relative to the full model, the retained effects are easier to interpret and the core conclusions are unchanged: HCC score, population quartile, and the age-by-political-affiliation pattern remain the main drivers in this state-level model.

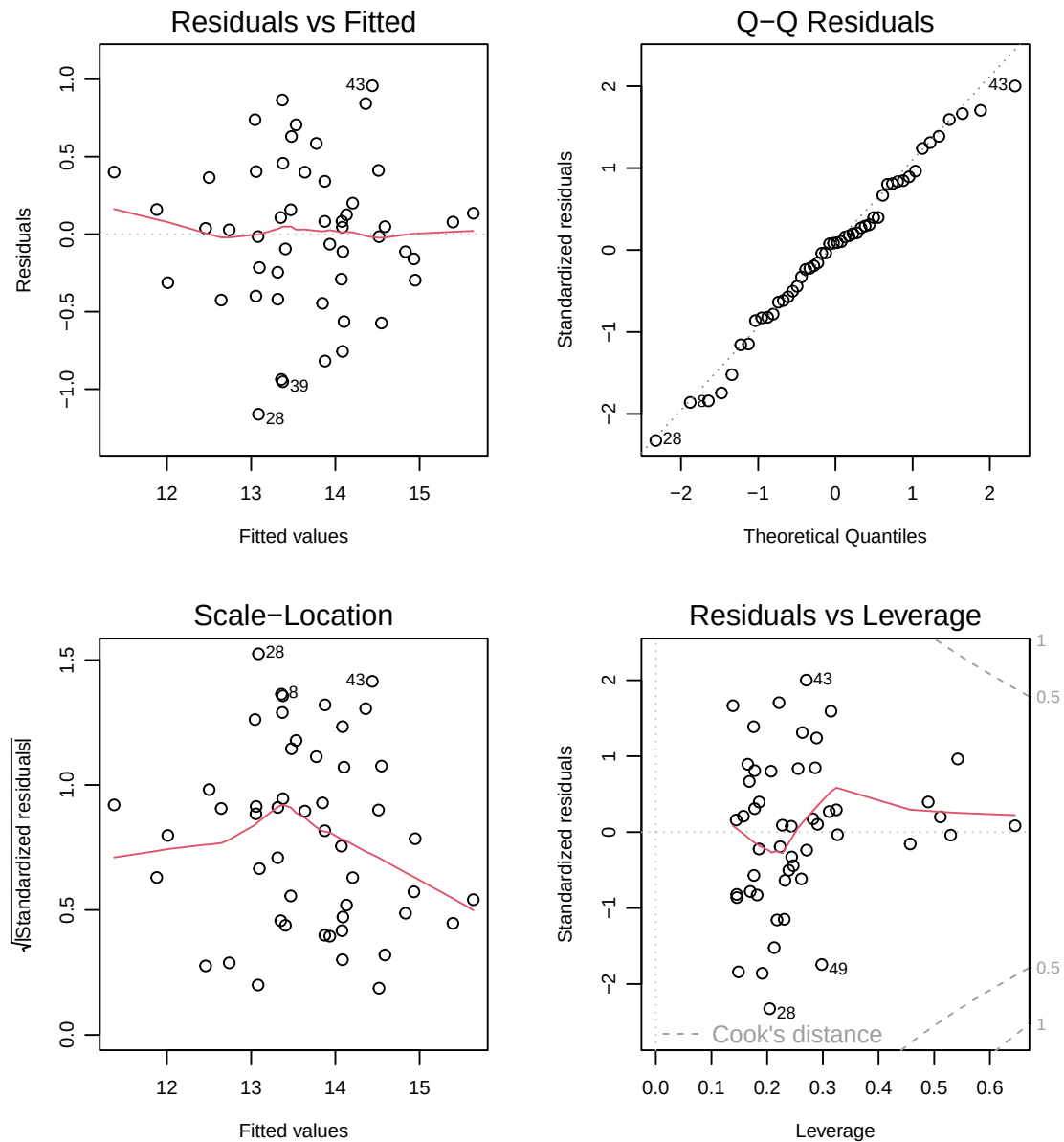
Term	VIF
Centered age	5.629
Republican	4.396
Age x Republican	4.152
Education Q4 High	2.890
Age x Swing	2.327
Education Q3	2.288
Swing	2.164
Population Q2	1.894
Population Q3	1.867
Population Q4 High	1.812
Education Q2	1.808
Centered HCC score	1.532

The VIF values show that centering helped keep the interaction model interpretable. The largest values occur for centered age and the Republican interaction terms, which is expected when an interaction is retained, but the values remain moderate rather than extreme.

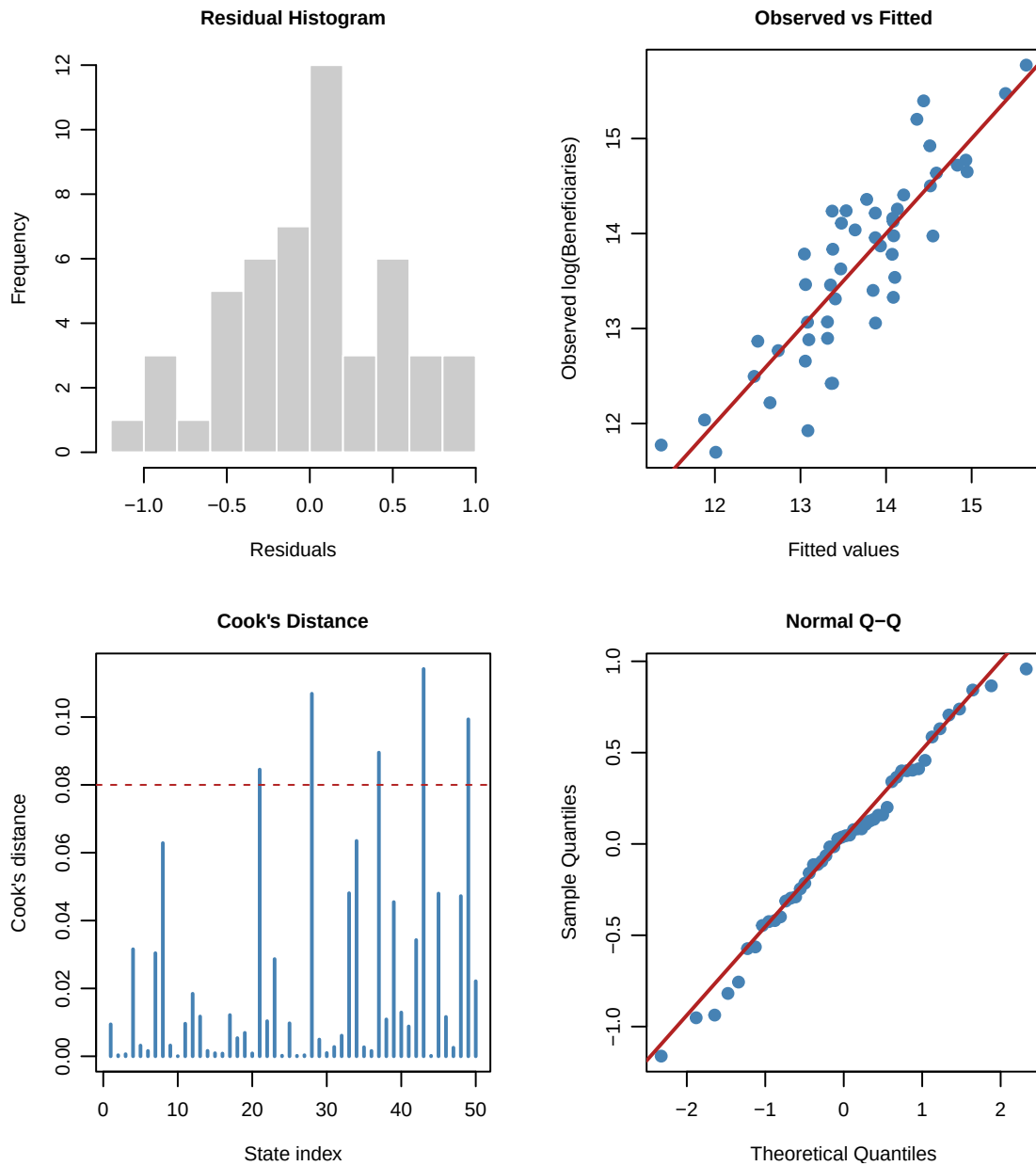
Check	Value
Residual standard error	0.56
Adjusted R-squared	0.679
Overall model p-value	<0.001
PRESS RMSE	0.627
Largest leverage	0.645
Largest studentized residual	2.482
Maximum Cook's distance	0.114

In particular, the small difference between residual standard error and PRESS RMSE suggests that the final model is not overfit relative to the size of the data set, while the leverage, studentized-residual, and Cook's-distance values give a compact screen for unusual or influential states.

Step 5: Diagnose the final model graphically



These standard regression diagnostics give the broad assumption check for the final model. The residual-versus-fitted and scale-location panels do not show a strong funnel pattern, the Q-Q plot is reasonably straight, and the residual-versus-leverage panel does not reveal a state that overwhelms the fit.



The final page provides a few more interpretable diagnostics beyond the standard `plot.lm()` output. The observed-versus-fitted panel shows that the model tracks the overall ordering of states reasonably well on the log scale, the residual histogram is close to symmetric, and the Cook's-distance panel confirms that the largest influence values remain well below a level that would suggest a single-state driven result.

Appendix E: Data Sources and Additional Help

Data Sources

- Age and sex. (2023). *American Community Survey*. <https://data.census.gov/chart/ACSST1Y2023.S0101?q=median+age+by+state&measure=LINE32&sort=abc>
- Electoral map: Blue or red states since 2000. (2026). *270toWin*. <https://www.270towin.com/states/>
- Medicare Geographic Variation - by National, State & County. (2025). *Centers for Medicare & Medicaid Services*. <https://data.cms.gov/summary-statistics-on-use-and-payments/medicare-geographic-comparisons/medicare-geographic-variation-by-national-state-county>
- Newitt, P. (2025). *Physicians per capita in all 50 states*. Becker's ASC Review. <https://www.beckersasc.com/asc-news/physicians-per-capita-in-all-50-states-2/>
- US Census Bureau. (2024, September 15). *Median annual family income in the United States in 2023, by state (in U.S. dollars)* [Graph]. In *Statista*. Retrieved April 12, 2026, from <https://www.statista.com/statistics/236804/median-family-income-in-the-united-states-by-state/>
- Erickson, W., Lee, C., & von Schrader, S. (2026). *Disability statistics from the American Community Survey (ACS)*. Cornell University Yang-Tan Institute. <https://www.disabilitystatistics.org/>

Additional Help

- OpenAI. (2026). *ChatGPT* [Large language model]. <https://chatgpt.com/>